

**CS + CSE Core**  
(16\*3 = 48 Credits)

|        |                                                                                                           |                 |                                               |
|--------|-----------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------|
| cse110 | <b>java 1: Introduction</b><br>Official Name: Programming Language 1                                      | <b>CS minor</b> | cse161 + cse162L,<br>eee/ece103 + eee/ece103L |
| cse111 | <b>java 2: Object Oriented Programming</b><br>Official Name: Programming Language 2                       | <b>CS minor</b> |                                               |
|        |                                                                                                           |                 |                                               |
| cse230 | <b>Discrete Mathematics</b>                                                                               | <b>CS minor</b> |                                               |
| cse260 | <b>Digital Logic Design</b>                                                                               | <b>CS minor</b> | eee/ece283 + eee/ece283L<br>eee301 + eee302   |
| cse340 | <b>RISC-V Assembly</b><br>Official Name: Computer Architecture                                            | <b>CS minor</b> | eee410                                        |
|        |                                                                                                           |                 |                                               |
| cse220 | <b>Algorithms 1</b><br>Official Name: Data Structures                                                     | <b>CS minor</b> |                                               |
| cse221 | <b>Algorithms 2</b><br>Official Name: Algorithm Analysis & Design                                         | <b>CS minor</b> |                                               |
| cse330 | <b>Algorithms 3: Numerics</b><br>Official Name: Numerical Methods                                         |                 |                                               |
| cse331 | <b>Algorithms 4: Logic, State-Machines &amp; Computability</b><br>Official Name: Automata & Computability |                 |                                               |
|        |                                                                                                           |                 |                                               |
| cse370 | <b>Database 1</b><br>Official Name: Database Systems                                                      | <b>CS minor</b> |                                               |
| cse321 | <b>Operating Systems</b>                                                                                  |                 |                                               |
| cse420 | <b>Compiler Design</b>                                                                                    |                 |                                               |
| cse421 | <b>Networking 2</b><br>Official Name: Computer Networks                                                   |                 | eee465 + eee466                               |
| cse470 | <b>Software Engineering</b>                                                                               | <b>CS minor</b> |                                               |
|        |                                                                                                           |                 |                                               |
| cse422 | <b>Artificial Intelligence</b>                                                                            |                 | eee472                                        |
| cse423 | <b>Computer Graphics</b>                                                                                  |                 |                                               |
|        |                                                                                                           |                 |                                               |

**CSE Core / (but) / CS Elective**  
(9\*3 = 27 Credits)

|        |                                                                         |  |                         |
|--------|-------------------------------------------------------------------------|--|-------------------------|
| cse250 | <b>Circuits 1</b><br>Official Name: Circuits and Electronics            |  |                         |
| cse251 | <b>Circuits 2</b><br>Official Name: Electronic Devices and Circuits     |  |                         |
| cse350 | <b>Circuits 3</b><br>Officially: Digital Electronics & Pulse Techniques |  |                         |
|        |                                                                         |  |                         |
| cse341 | <b>Microprocessors</b>                                                  |  |                         |
| cse360 | <b>Computer Interfacing</b>                                             |  |                         |
| cse460 | <b>VLSI</b>                                                             |  |                         |
| cse461 | <b>Robotics</b>                                                         |  |                         |
|        |                                                                         |  |                         |
| cse320 | <b>Networking 1</b><br>Official Name: Data Communications               |  | eee/ece361 + eee/ece362 |
| cse471 | <b>Systems Analysis &amp; Design</b>                                    |  |                         |

## CSE + CS Elective

( CSE: 6 Credits - CS: 21 Credits )

|         |                                        | su26 Activity | Source:                    |
|---------|----------------------------------------|---------------|----------------------------|
| cse310  | OOP                                    | not Active    |                            |
| cse342  | Computer System Engineering            | not Active    |                            |
| cse462  | Fault Tolerance System                 | not Active    |                            |
|         |                                        |               |                            |
|         |                                        |               |                            |
| cse390  | Technical Communication                | not Active    |                            |
| cse391  | Programming for the Internet           | Active        |                            |
| cse410  | Advanced Programming in UNIX           | not Active    |                            |
| cse473  | Financial Engineering                  | not Active    |                            |
| cse483  | Software Engineering II                | not Active    |                            |
| cse489  | Android App Development                | not Active    |                            |
| cse492  | iOS App Development                    | not Active    |                            |
| cse493  | Enterprise and Software Architecture   | not Active    |                            |
| cse495  | Internet of Things (IoT)               | not Active    |                            |
| cse499  | E-commerce                             | not Active    |                            |
| cse586  | Advanced Web Development               | not Active    |                            |
| cse490A | User Engagement / Gamification         | Active        |                            |
|         |                                        |               |                            |
|         |                                        |               |                            |
| cse451  | Network Science and Models             | not Active    |                            |
| cse484  | Cloud Computing                        | not Active    |                            |
| cse485  | Advanced Cloud Computing               | not Active    |                            |
| cse486  | WAN Routing and Technologies           | not Active    |                            |
| cse487  | Web Services                           | not Active    |                            |
| cse488  | Serverless and Microservices           | not Active    |                            |
|         |                                        |               |                            |
|         |                                        |               |                            |
| cse458  | Quantum Complexity Theory              | not Active    |                            |
| cse481  | Quantum Computing I                    | Active        |                            |
| cse482  | Quantum Computing II                   | not Active    |                            |
| cse490C | Quantum Computing III                  | not Active    |                            |
|         |                                        |               |                            |
| cse401  | Concrete Mathematics                   |               |                            |
| cse402  | Optimization                           | Active        |                            |
| cse419  | codeforces 1                           | not Active    |                            |
| cse426  | Advanced Algorithms                    | Active        |                            |
| cse449  | Parallel, Distributed, & HPC           | Active        | High-Performance Computing |
| cse450  | codeforces 2                           | not Active    |                            |
| cse452  | Computability and Complexity Theory    | not Active    |                            |
| cse453  | Theory of Computation II               | not Active    |                            |
| cse454  | Advanced Complexity Theory             | not Active    |                            |
| cse455  | Topics in Theoretical Computer Science | not Active    |                            |
| cse456  | Design and Analysis of Algorithms      | not Active    |                            |
| cse457  | Randomized Algorithms                  | not Active    |                            |
| cse474  | Simulation & Modeling                  | not Active    |                            |

|         |                                                    |            |                                |
|---------|----------------------------------------------------|------------|--------------------------------|
|         |                                                    |            |                                |
| cse472  | Human Computer Interface (HCI)                     | Active     |                                |
| cse494  | Advanced Robotics                                  | not Active |                                |
| cse498  | Advanced Graphics                                  | not Active |                                |
|         |                                                    |            |                                |
| cse392  | Signals & Systems                                  | not Active |                                |
| cse428  | Image Processing                                   | Active     | eee476                         |
| cse429  | Basic Multimedia Theory                            | not Active |                                |
| cse430  | Signal Processing                                  | Active     |                                |
|         |                                                    |            |                                |
|         |                                                    |            |                                |
| cse424  | Pattern Recognition                                | Active     |                                |
| cse425  | Neural Networks                                    | Active     | eee474                         |
| cse427  | Machine Learning                                   | Active     |                                |
| cse431  | NLP                                                | not Active | alternatives = cse442 / cse439 |
| cse432  | Speech Recognition & Synthesis                     | not Active |                                |
| cse433  | ML for healthcare                                  | not Active |                                |
| cse434  | ML for finance                                     | not Active |                                |
| cse435  | Reinforced Learning                                | not Active |                                |
| cse439  | Machine Vision                                     | not Active |                                |
| cse440  | NLP 2                                              | Active     | Natural Language Processing    |
| cse441  | Machine Learning II                                | not Active |                                |
| cse442  | NLP Driven UI/UX                                   | not Active |                                |
| cse463  | Computer Vision 1                                  | Active     |                                |
|         |                                                    |            |                                |
|         |                                                    |            |                                |
| cse436  | Data Visualization                                 | not Active |                                |
| cse437  | Data Science: Coding with Real world Data          | Active     |                                |
| cse438  | Data Science: Applied                              | not Active |                                |
| cse445  | Statistical Data Analysis                          | not Active |                                |
| cse490D | Football Data & Analytics                          | Active     |                                |
|         |                                                    |            |                                |
| cse443  | Bio-Informatics 1                                  | Active     |                                |
| cse444  | Bio-Informatics 2                                  | not Active |                                |
| cse459  | Introduction to Computational Molecular Biology    | not Active |                                |
| cse497  | Computational Cognitive Science                    | not Active |                                |
|         |                                                    |            |                                |
| cse446  | Blockchain and Cryptocurrencies                    | Active     |                                |
| cse447  | Cryptography and Cryptanalysis                     | Active     |                                |
| cse496  | Ethical Hacking                                    | not Active |                                |
| cse490B | Intro to CyberSecurity                             | Active     |                                |
|         |                                                    |            |                                |
|         |                                                    |            |                                |
| cse500  | Ethics of Data in Social Context                   | not Active |                                |
| cse502  | Scientific & Engineering Innovation                | not Active |                                |
| cse503  | Project Engineering                                | not Active |                                |
| cse504  | Workshop for Entrepreneurs and Innovators          | not Active |                                |
| cse505  | Patents, Copyrights, and Intellectual Property Law | not Active |                                |
| cse506  | Ethics                                             | not Active |                                |
| cse507  | Introduction to Design Thinking and Innovation     | not Active |                                |
|         |                                                    |            |                                |
|         |                                                    |            |                                |
|         |                                                    |            |                                |

|                                          |                                                                      |                           |                                  |
|------------------------------------------|----------------------------------------------------------------------|---------------------------|----------------------------------|
| cse400                                   | Project & Thesis                                                     | 4 Credits                 |                                  |
| <b>School Core</b><br>(4*3 = 12 Credits) |                                                                      |                           |                                  |
| mat120                                   | MATH 2: Integration & Differential Equations                         |                           |                                  |
| mat216                                   | MATH 3: Linear Algebra & Fourier Analysis                            |                           |                                  |
| mat215                                   | MATH 4: Complex Variables & Laplace Transformations                  |                           |                                  |
| phy112                                   | Principles of Physics                                                |                           |                                  |
|                                          |                                                                      |                           |                                  |
| <b>GenEd</b><br>(13*3 = 39 Credits)      |                                                                      |                           |                                  |
| ENG091                                   | Foundation Course<br>(0 Credit)                                      | 2<br>(without eng091)     | 1. Writing<br>Comprehension      |
| ENG101                                   | English Fundamentals                                                 |                           |                                  |
| ENG102                                   | Composition I                                                        |                           |                                  |
| ENG103                                   | Advanced Writing Skills & Presentation<br>(only for ENG102 freshers) |                           |                                  |
|                                          |                                                                      |                           |                                  |
|                                          |                                                                      |                           |                                  |
| BIO101                                   | Introduction to Biology                                              | 2                         | 2. Math<br>&<br>Natural Sciences |
| CHE101                                   | Introduction to Chemistry                                            |                           |                                  |
| CSE110                                   | Introduction to Computer Science<br>(alt: cse101)                    |                           |                                  |
| ENV103                                   | Elements of Environmental Science                                    |                           |                                  |
| MAT101                                   | Fundamentals of Mathematics                                          |                           |                                  |
| MAT110                                   | MATH I: Differential Calculus and Co-ordinate Geometry               |                           |                                  |
| PHY101                                   | Introduction to Physics                                              |                           |                                  |
| PHY111                                   | Principles of Physics I                                              |                           |                                  |
| STA101                                   | Introduction to Statistics                                           |                           |                                  |
| STA201                                   | Elements of Statistics and Probabilities                             |                           |                                  |
| GSC110                                   | Space, Time & Infinity                                               |                           |                                  |
|                                          |                                                                      |                           |                                  |
|                                          |                                                                      |                           |                                  |
| BNG103                                   | Bangla Language and Literature                                       | bng103 +<br>hum103 +<br>1 | 3. Arts &<br>Humanities          |
| ENG113                                   | Introduction to English Poetry                                       |                           |                                  |
| ENG114                                   | Introduction to English Drama                                        |                           |                                  |
| ENG115                                   | Introduction English Prose                                           |                           |                                  |
| ENG333                                   | Globalization and the Media                                          |                           |                                  |
| HST102                                   | The Modern World                                                     |                           |                                  |
| HST103                                   | History of Bangladesh                                                |                           |                                  |
| HST104                                   | Global History Lab - A History of World since 1300                   |                           |                                  |
| HUM101                                   | World Civilization & Culture                                         |                           |                                  |
| HUM102                                   | Introduction to Philosophy<br>(alt: hum112)                          |                           |                                  |
| HUM103                                   | Ethics and Culture                                                   |                           |                                  |
| HUM 207                                  | Narratives of Truth and Lies                                         |                           |                                  |
|                                          |                                                                      |                           |                                  |
|                                          |                                                                      |                           |                                  |
|                                          |                                                                      |                           |                                  |
| ANT101                                   | Introduction to Anthropology                                         |                           |                                  |
| ANT342                                   | Body and Society                                                     |                           |                                  |
| ANT351                                   | Gender & Development                                                 |                           |                                  |
| BUS102                                   | Business - Basics, Ethics and Environment                            |                           |                                  |
| BUS201                                   | Business and Human Communication                                     |                           |                                  |
| BUS333                                   | Social Entrepreneurship Practicum                                    |                           |                                  |

|                                                                                            |                                                                                              |                                 |                                           |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------|
| BUS334                                                                                     | Social Intrapreneurship Practicum                                                            | emb101<br>/dev101 +<br><b>1</b> | 4. Social Sciences                        |
| BUS335                                                                                     | Sustainable Development & Social Enterprise                                                  |                                 |                                           |
| DEV104                                                                                     | Foundations of International Development                                                     |                                 |                                           |
| DEV201                                                                                     | Health, Culture & Development                                                                |                                 |                                           |
| ECO101                                                                                     | Introduction to Microeconomics                                                               |                                 |                                           |
| ECO102                                                                                     | Introduction to Macroeconomics                                                               |                                 |                                           |
| ECO105                                                                                     | Introduction to Economics                                                                    |                                 |                                           |
| EMB101                                                                                     | Emergence of Bangladesh                                                                      |                                 |                                           |
| POL101                                                                                     | Introduction to Political Science                                                            |                                 |                                           |
| POL102                                                                                     | Comparative Governance                                                                       |                                 |                                           |
| POL103                                                                                     | International Relations and Global Politics                                                  |                                 |                                           |
| POL201                                                                                     | Introduction to Civic Engagement                                                             |                                 |                                           |
| POL202                                                                                     | Foundations in Public Policy                                                                 |                                 |                                           |
| POL203                                                                                     | Political Theory                                                                             |                                 |                                           |
| POL210                                                                                     | Youth & Global Movement                                                                      |                                 |                                           |
| PSY101                                                                                     | Introduction to Psychology                                                                   |                                 |                                           |
| SOC101                                                                                     | Introduction to Sociology                                                                    |                                 |                                           |
| SOC201                                                                                     | Social Inequality<br>(alt: ant202)                                                           |                                 |                                           |
| BU201                                                                                      | Community Engagement & Action                                                                |                                 |                                           |
|                                                                                            |                                                                                              |                                 |                                           |
|                                                                                            |                                                                                              |                                 |                                           |
| CST201                                                                                     | Luxury & Oppression in a Globalized World                                                    | 1                               | 5. Communities,<br>Seeking Transformation |
| CST204                                                                                     | Communication for Social Change                                                              |                                 |                                           |
| CST301                                                                                     | For the Love of Food                                                                         |                                 |                                           |
| CST302                                                                                     | Communication for Social Change                                                              |                                 |                                           |
| CST303                                                                                     | Law for Life, Peace and Justice                                                              |                                 |                                           |
| CST304                                                                                     | Documentary Film: Theory and Practice                                                        |                                 |                                           |
| CST305                                                                                     | Borders and Beyond: Past and Future                                                          |                                 |                                           |
| CST306                                                                                     | Ethical Leadership                                                                           |                                 |                                           |
| CST307                                                                                     | Art, Community and the Future                                                                |                                 |                                           |
| CST308                                                                                     | Social Dimensions of Faith and Development                                                   |                                 |                                           |
| CST309                                                                                     | Global Citizenship                                                                           |                                 |                                           |
| CST310                                                                                     | Social Cohesion and Peace Building                                                           |                                 |                                           |
| CST314                                                                                     | Visual Storytelling: Documentary Film                                                        |                                 |                                           |
| CST333                                                                                     | Personal Finance for Sustainable Economic Wellbeing                                          |                                 |                                           |
|                                                                                            |                                                                                              |                                 |                                           |
| Total: 10*3 = 30 Credits                                                                   |                                                                                              |                                 |                                           |
|                                                                                            | Where did 3 courses vanish?                                                                  |                                 |                                           |
|                                                                                            | --> You have to take 3 courses outside your department.<br>MNS/MPS/EEE/GenEd Anything works. |                                 |                                           |
|                                                                                            |                                                                                              |                                 |                                           |
| Students enrolled in and after Fall 20 will have to complete 39 credits (13 GenEd courses) |                                                                                              |                                 |                                           |
| Students enrolled before Fall 20 will have to complete 42 credits                          |                                                                                              |                                 |                                           |
|                                                                                            |                                                                                              |                                 |                                           |
|                                                                                            |                                                                                              |                                 |                                           |
|                                                                                            |                                                                                              |                                 |                                           |
|                                                                                            |                                                                                              |                                 |                                           |

| Post Graduate (cse700) |                                         |  |  |
|------------------------|-----------------------------------------|--|--|
| cse705                 | Computer Communication and Networks     |  |  |
| cse706                 | Parallel Algorithms                     |  |  |
| cse707                 | Distributed Computing Systems           |  |  |
| cse708                 | Graph Theory                            |  |  |
| cse710                 | Advanced Artificial Intelligence        |  |  |
| cse711                 | Symbolic Machine Learning I             |  |  |
| cse712                 | Symbolic Machine Learning II            |  |  |
| cse713                 | Advanced Syntactic Pattern Recognition  |  |  |
| cse714                 | Speech Recognition                      |  |  |
| cse715                 | Neural Network and Fuzzy Systems        |  |  |
| cse716                 | Advanced Database Sytems                |  |  |
| cse717                 | Mathematical Programming                |  |  |
| cse718                 | Petri-Net Theory & Modeling of Systems  |  |  |
| cse719                 | Multimedia Systems                      |  |  |
| cse720                 | Computer Graphics and Animation         |  |  |
| cse721                 | Introduction to Cryptography            |  |  |
| cse722                 | Applied Cryptography & Protocols        |  |  |
| cse723                 | Web Security                            |  |  |
| cse724                 | Ethical Hacking                         |  |  |
| cse725                 | CyberSecurity Policy & Risk Management  |  |  |
| cse726                 | Information Assurance                   |  |  |
| cse727                 | Digital Forensics                       |  |  |
| cse728                 | Privacy Enhancing Technologies          |  |  |
| cse729                 | Blockchain & Distributed Ledger         |  |  |
| cse730                 | Threat Intelligence & Incident Response |  |  |
| cse731                 | HCI & Design                            |  |  |
| cse732                 | Advanced Quantum Computing              |  |  |
| cse741                 | Advanced Cloud Computing                |  |  |
| cse751                 | Advanced NLP                            |  |  |
| cse752                 | Deep Neural Networks                    |  |  |
| cse753                 | Advanced Reinforced Learning            |  |  |
| cse754                 | Deep Learning for Computer Vision       |  |  |
| cse755                 | Generative Machine Learning             |  |  |
| cse756                 | Probabilistic Graphical Models          |  |  |
| cse757                 | Machine Learning Systems Design         |  |  |
| cse758                 | Data Mining                             |  |  |
| cse759                 | AI in Healthcare & Medicine             |  |  |
| cse761                 | Big Data Analytics                      |  |  |
| cse762                 | Fairness, Ethics, Explainablity in AI   |  |  |
| cse763                 | Advanced Bioinformatics                 |  |  |
| cse771                 | Economics of CyberSecurity              |  |  |
| cse772                 | Secure Programming                      |  |  |
| cse773                 | Usable Security & Privacy               |  |  |
| cse774                 | Malware Analysis                        |  |  |
| cse791                 | Research Methodology & Seminar          |  |  |
| cse798                 | Project                                 |  |  |
| cse799A                | Gamification                            |  |  |
| cse799                 | Decision Support System                 |  |  |